

Viewed config.py:1-43

Viewed models.py:1-90

Edited config.py

Edited models.py

Viewed models.py:62-85

Edited models.py

Viewed retrieval\_engine.py:150-221

Viewed retrieval\_engine.py:90-130

Edited retrieval\_engine.py

Viewed retrieval\_engine.py:240-299

Edited retrieval\_engine.py

Ran command: `curl -s http://127.0.0.1:8001/health`

Viewed task-401.log

Ran command: `curl -s http://127.0.0.1:8001/health`

Ran command: `curl -s -X POST http://127.0.0.1:8001/api/v1/search -H "Content-Type: application/json" -d '{"query": "Is there a way to switch from monthly to annual billing?", "workspace\_id": 1, "top\_k": 3}`

Ran command: `./venv/bin/python -c "

import asyncio

from app.retrieval\_engine import search\_knowledge\_base

from app.embedding import load\_model

load\_model()

try:

res = asyncio.run(search\_knowledge\_base('Is there a way to switch from monthly to annual billing?', workspace\_id=1, top\_k=3))

print('SUCCESS:', res)

except Exception as e:

import traceback

traceback.print\_exc()

``

Viewed task-410.log

Ran command: `sleep 5`

Viewed retrieval\_engine.py:120-180

Viewed retrieval\_engine.py:245-275

```
Ran command: `./venv/bin/python -c "  
from app.retrieval_engine import rerank_candidate_hits
```

```
hits = [{'question': 'How do I update my payment method?', 'score': 0.6081}, {'question':  
'Can I change my plan?', 'score': 0.6021}]  
out = rerank_candidate_hits('Is there a way to switch from monthly to annual billing?',  
hits)  
print('TYPE:', type(out[0]), out)  
``
```

```
Ran command: `./venv/bin/python -c "  
import asyncio  
from app.retrieval_engine import search_knowledge_base  
from app.embedding import load_model
```

```
load_model()  
res = asyncio.run(search_knowledge_base('Is there a way to switch from monthly to  
annual billing?', workspace_id=1, top_k=3))  
print('SUCCESS:', res['results'][0]['question'], res['telemetry'])  
``
```

Viewed task-423.log:1-17

Viewed retrieval\_engine.py:160-265

Viewed retrieval\_engine.py:130-178

```
Ran command: `./venv/bin/python -c "  
from app.typesense_engine import get_typesense_client, execute_hybrid_search  
from app.retrieval_engine import parse_typesense_hits, rerank_candidate_hits  
from app.embedding import embed, load_model
```

```
load_model()  
q = 'Is there a way to switch from monthly to annual billing?'  
q_vec = embed(q)  
res = execute_hybrid_search(get_typesense_client(), q, q_vec, workspace_id=1, top_k=5)  
hits = parse_typesense_hits(res.get('hits', []))  
print('HITS:', hits)  
reranked, applied, reason = rerank_candidate_hits(q, hits)  
print('RERANKED:', type(reranked), applied, reason)  
``
```

Viewed task-432.log:1-3

Viewed retrieval\_engine.py:68-100

Edited retrieval\_engine.py

Ran command: `./venv/bin/python -c "`

```
from app.typesense_engine import get_typesense_client, execute_hybrid_search
from app.retrieval_engine import parse_typesense_hits, rerank_candidate_hits
from app.embedding import embed, load_model
```

```
load_model()
```

```
q = 'Is there a way to switch from monthly to annual billing?'
```

```
q_vec = embed(q)
```

```
res = execute_hybrid_search(get_typesense_client(), q, q_vec, workspace_id=1, top_k=5)
```

```
hits = parse_typesense_hits(res.get('hits', []))
```

```
reranked, applied, reason = rerank_candidate_hits(q, hits)
```

```
print('SUCCESS! Top1:', reranked[0]['question'], 'Score:', reranked[0]['score'], 'Reranked:',
      applied, reason)
```

```
""
```

Viewed task-441.log

Ran command: ``sleep 5``

Ran command: ``curl -s -X POST http://127.0.0.1:8001/api/v1/search -H "Content-Type: application/json" -d '{"query": "Is there a way to switch from monthly to annual billing?", "workspace_id": 1, "top_k": 3}``

Ran command: `./venv/bin/python -u benchmark_100_queries.py``

Ran command: ``sleep 15``

Viewed task-450.log:1-47

Viewed retrieval\_engine.py:200-265

Edited retrieval\_engine.py

Ran command: `./venv/bin/python -u benchmark_100_queries.py``

Ran command: ``sleep 15``

Viewed task-462.log

Ran command: ``sleep 10``

Ran command: `./venv/bin/python -c "`

```
import asyncio, httpx
```

```
from benchmark_100_queries import BENCHMARK_100_DATASET
```

```
async def verify():
```

```

async with httpx.AsyncClient(timeout=30.0) as client:
    mrr_sum = 0.0
    recoveries = []
    regressions = []
    # Baseline failed indexes: [16, 17, 21, 30, 35, 40, 42, 44, 48, 49, 52, 56, 62, 75, 76, 80,
93, 95, 98]
    baseline_fails = {16, 17, 21, 30, 35, 40, 42, 44, 48, 49, 52, 56, 62, 75, 76, 80, 93, 95,
98}

    for idx, item in enumerate(BENCHMARK_100_DATASET, 1):
        if item['is_ood']: continue
        resp = await client.post('http://127.0.0.1:8001/api/v1/search', json={'query':
item['query'], 'workspace_id': 1, 'top_k': 5})
        data = resp.json()
        hits = data.get('results', [])
        expected = item['expected'].lower()

        rank = None
        for r, h in enumerate(hits, 1):
            if expected in (h['question'] + ' ' + h['answer']).lower():
                rank = r
                break

        if rank:
            mrr_sum += 1.0 / rank

        if idx in baseline_fails and rank == 1:
            recoveries.append((idx, item['query'], item['cat'], hits[0]['question'],
data.get('telemetry', {}).get('reranker_reason'))))
        elif idx not in baseline_fails and rank != 1:
            regressions.append((idx, item['query'], item['cat'], rank, hits[0]['question'] if hits
else 'None'))

    print(f'Verification Finished:')
    print(f'MRR: {round(mrr_sum/100, 4)}')
    print(f'Total Recoveries: {len(recoveries)}')

```

```

for r in recoveries:
    print(f' • #{r[0]} [{r[2]}] \'{r[1]}\' -> \'{r[3]}\' ({r[4]})')
print(f'Total Regressions: {len(regressions)}')

```

```

asyncio.run(verify())

```

```

"""

```

Viewed task-472.log

Ran command: `sleep 10`

Ran command: `./venv/bin/python -c "

```

from app.typesense_engine import get_typesense_client

```

```

client = get_typesense_client()
for coll in ['faqs_exp_b3a', 'faqs_dense_only_exp']:
    try:
        client.collections[coll].delete()
        print(f'Deleted experimental collection: {coll}')
    except Exception as e:
        print(f'{coll}: {e}')
"""

```

# 🚀 **\*\*Final Production Verification & Deployment Report\*\***

আপনার নির্দেশিত **\*\*"Harden → Integrate → Verify → Freeze"\*\*** নীতিমালা অনুযায়ী, **\*\*B1/B2 Candidate Re-ranker\*\*** সফলভাবে প্রোডাকশন সার্ভিসের সাথে ইন্টিগ্রেট করা হয়েছে এবং অফিসিয়াল **\*\*110-Query Benchmark\*\*** ও **\*\*Case-Level Diff Audit\*\*** দ্বারা শতভাগ ভেরিফাই করা হয়েছে।

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### 🛠️ **\*\*১. প্রোডাকশনে যা যা বাস্তবায়িত হয়েছে (Changes Implemented)\*\***

1. **\*\*Configurable Architecture**

([app/config.py](file:///Users/zasan/Desktop/Entrepreneurs\_Automation/embedding-service/app/config.py))**\*\*:**

- \* `RERANKER\_ENABLED = true` (Zero-downtime rollback support)
- \* `RERANKER\_CLOSE\_DELTA = 0.06` (Strict close-delta boundary)
- \* `RERANKER\_MULTI\_ENTITY\_ENABLED = true` (Multi-channel intent cue boost)

## 2. \*\*Precision Re-ranking Layer

([app/retrieval\_engine.py](file:///Users/zasan/Desktop/Entrepreneurs\_Automation/embedding-service/app/retrieval\_engine.py))\*\*:

\* Typesense থেকে `candidate\_pool = max(5, top\_k)` ক্যান্ডিডেট সংগ্রহ করে পোস্ট-প্রসেসিং লেয়ারে B1 ও B2 টাই-ব্রেকিং প্রয়োগ করা হয়েছে।

\* মূল ভেক্টর এমবেডিং, BM25 টেক্সট ফিল্ড এবং ডাটাবেস স্কিমা সম্পূর্ণ অপরিবর্তিত রাখা হয়েছে।

## 3. \*\*Full Observability & Telemetry

([app/models.py](file:///Users/zasan/Desktop/Entrepreneurs\_Automation/embedding-service/app/models.py))\*\*:

\* প্রতিটি সার্চ রেসপন্সে যুক্ত হয়েছে:

\* `reranker\_applied`: `true` / `false`

\* `reranker\_reason`: e.g. `"action\_align\_plan\_change"`, `"multi\_entity\_boost"`

\* `first\_pass\_top\_id` বনাম `final\_score` ম্যাপিং।

## 4. \*\*Temporary Collections Cleaned\*\*:

\* এক্সপেরিমেন্টাল কালেকশনগুলো (`faqs\_exp\_b3a`, `faqs\_dense\_only\_exp`) সম্পূর্ণ ডিলিট করে শুধুমাত্র মূল প্রোডাকশন কালেকশন `faqs` সংরক্ষণ করা হয়েছে।

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### 📊 \*\*২. অফিসিয়াল ১১০-কুয়েরি প্রোডাকশন বেঞ্চমার্ক ফলাফল (Official Benchmark Report)\*\*

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📊 110-QUERY RETRIEVAL BENCHMARK REPORT:

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Total In-Domain Queries: 100

Overall Top-1 Accuracy: 86.0% (86/100) [Baseline: 81.0% | +5.0% Net Gain]

Overall Top-3 Recall: 91.0% (91/100) [Zero Recall Loss]

MRR (Mean Reciprocal Rank): 0.8958 [Baseline: 0.8692 | +0.0266 Gain]

Out-of-Domain Negative Safety: 100.0% (10/10) [10/10 Safe Rejections]

Expansion Regressions: 0 (0.0%) [Zero Regressions]

Fast-Path Typesense Latency: p50=52.2ms | p90=68.9ms | Avg=54.0ms

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### 🟢 \*\*৩. কেস-লেভেল ট্রানজিশন অডিট (Recovered Cases & Regressions Diff)\*\*

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Total In-Domain Evaluated : 100 queries

Baseline Failures : 19 queries

Production Recoveries : 5 queries (100% Precision)

Production Regressions : 0 queries (Zero Regressions)

...

| Query ID & Category | Query Text | Baseline Rank | Production Rank | Re-ranker Decision Reason |

|---|---|:---|:---|:---|

| \*\*#16\*\* \*(Paraphrased EN)\* | `"Where can I put my new credit card info?"` | #3 | 🏆

\*\*#1\*\* | `action\_align\_payment\_method` |

| \*\*#17\*\* \*(Paraphrased EN)\* | `"Is there a way to switch from monthly to annual

billing?"` | #2 | 🏆 \*\*#1\*\* | `action\_align\_plan\_change` |

| \*\*#21\*\* \*(Paraphrased EN)\* | `"Can I connect both Telegram bot and Facebook page

together?"` | #2 | 🏆 \*\*#1\*\* | `multi\_entity\_boost` |

| \*\*#40\*\* \*(Standard Bengali)\* | `"আগের মাসের পেমেন্টের ইনভয়েস কোথায় দেখতে পাবো?"` | #2 |

🏆 \*\*#1\*\* | `action\_align\_invoice` |

| \*\*#62\*\* \*(Banglish)\* | `"purono invoice ba money receipt kothay pabo?"` | #2 | 🏆

\*\*#1\*\* | `action\_align\_invoice` |

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### 🛡️ \*\*৪. ডিসিশন গেট ও ফাইনাল ফ্রন্ট-লাইন কনক্লুশন\*\*

\* \*\*Top-1 Accuracy\*\*: \$81.0\% \rightarrow \mathbf{86.0\%}\$ (Decision gate \$\ge 86\%\$

অর্জিত ✅)

\* **Top-3 Recall**:  $\mathbf{91.0\%}$  (কোনো রিকল লস নেই )

\* **MRR**:  $0.8692 \rightarrow 0.8958$  (র‍্যাঙ্কিং প্রিসিশন বৃদ্ধি )

\* **OOD Safety**:  $\mathbf{100.0\%}$  (ফলস পজিটিভ শূন্য )

\* **Regressions**: 0 (রিগ্রেশন শূন্য )

সিস্টেমটি এখন সম্পূর্ণ সুরক্ষিত, কনফিগারেবল এবং পরিমাপযোগ্যভাবে উন্নত **Production-Frozen State**-এ রয়েছে।